

Implications of Physiological Stress from Exposure to Heavy Metals in *Hemigrapsus nudus*

Abstract

Heavy metal contamination is a widespread environmental stressor in coastal ecosystems, where intertidal crustaceans are often exposed through water, sediment, and food sources. Copper and lead are of particular concern due to their ability to bioaccumulate and disrupt physiological processes, even at sublethal concentrations (Amado et al. 2012; Rainbow 2002). The purple shore crab, *Hemigrapsus nudus*, is a common intertidal species in the Pacific Northwest that inhabits nearshore environments frequently impacted by urban runoff, wastewater discharge, and shipping port activity. This study examined the physiological effects of copper and lead exposure on *H. nudus*. Over three weeks, individuals were exposed to these metals and assessed for changes in righting time, osmolality, and total hemocyte count (THC). Lead exposure significantly affected osmoregulation, indicated by increases in osmolality, while THC and righting time showed no statistically significant differences. These findings suggest that lead impairs the crabs' ability to maintain osmotic balance. Although copper exposure was associated with increased osmolality, the effect was not statistically significant, which may indicate either a low level of disruption or a greater capacity for physiological compensation, potentially due to copper's role as an essential ion in hemocyanin oxygen transport. Despite the lack of significant changes in THC and righting time, observed behavioral alterations in both treatment groups suggest a generalized stress response. These results further support the damage that heavy metal pollution has on intertidal regions, and should be used for future research on physiological impacts on crustaceans and other invertebrates.

Introduction

The Washington State Department of Health states that lead exposure can cause long-term harm and is especially toxic to children, with studies linking lead exposure to behavioral challenges. In Washington state alone, more than 2,500 children under the age of 6 tested between 2019 and 2023 had blood lead levels at or above 5 µg/dL (Washington State Department of Health). Heavy metal contamination is a widespread environmental stressor in coastal ecosystems, where intertidal crustaceans are often exposed through water, sediment, and food sources. Among these contaminants, copper and lead are of concern because they can accumulate in aquatic and terrestrial organisms and impact essential physiological processes, even at non-lethal concentrations (Amado et al. 2012; Rainbow 2002).

Crustaceans are useful bioindicators because of how their physiology integrates environmental change through measurable responses in immunity, osmoregulation, and their behavior (Adeleke et al. 2021). The purple shore crab, *Hemigrapsus nudus*, is a common intertidal species located in the Pacific Northwest that lives in nearshore habitats, often where metals are introduced into the ocean from urban runoff, sewage, and shipping ports. *H. nudus* are osmoregulators, and previous studies have shown that their hemolymph osmolality stays

relatively stable across exposure to a broad range of salinities, indicating that their internal water and ion balance is closely tied to environmental stress (Adeleke et al. 2021; Lignot et al. 2000).

Copper exposure in crustaceans is known to cause multiple physiological disruptions, including oxidative stress and altered immune responses (Truscott and White 1990). Although copper is an essential trace element and a component of hemocyanin (hemolymph), excessive concentrations can still be toxic (Coates and Costa-Paiva 2020). Lead is a nonessential metal known to affect reproductive and oxidative mechanisms, and has a stronger effect in weak osmoregulators than osmoconformers (Li et al. 2015; Amado et al. 2012; Rainbow 2002). When crustaceans are repeatedly exposed to environmental stress, they tend to show disrupted osmoregulatory capacity and immunological response. In a previous study, researchers found that the survival rate of sperm decreased after lead exposure in freshwater crabs as the accumulation of lead concentrated in their bodies was statistically significant (Li et al. 2015). In another, researchers found that stress often results in a decrease in ionic regulation originating in cellular alterations of Na⁺/K⁺-ATPase activity (Lignot et al. 2000). Heavy metals also affect hemocytes, the cells that manage innate immunity. Changes in total hemocyte count (THC) may reflect either immune activation or suppression, depending on the dose and length of exposure (Yildiz and Atar 2002).

The objective of this paper is to determine how the presence of heavy metals, specifically copper and lead, affects *Hemigrapsus nudus* physiology. Over the course of three weeks, a sample of *H. nudus* were exposed to copper and lead and tested for changes in righting time, osmolality, and total hemocyte count. The copper and lead experimental groups were expected to create sublethal stress that may impair immune function, slow behavioral recovery, and disrupt osmoregulation in *H. nudus*.

Methods

This experiment used 14 randomly selected *Hemigrapsus nudus* spread across 3 treatment groups: control, lead, and copper. Separate containers were used for each group, kept at ambient water temperatures and salinity, and a piece of lead and copper of approximately the same weight (lead: 29.18 g; copper: 30.22 g) were placed in their respective containers. 4 crabs were placed in each container, with the extra 2 crabs being used to extract baseline hemolymph. Each crab was then fitted with a B-tag, and we recorded tag color and number, weight, carapace length and width, and sex before collecting righting times.

Over the course of this experiment, the crabs were not fed, effectively removing a factor that could influence results. During week 2, two crabs from each treatment were randomly selected for righting time and hemolymph extraction. Due to mortalities between weeks 2 and 3, all remaining crabs were sampled on the final day to maximize available data.

Hemolymph was extracted using U-100 insulin syringes and inserted into the soft joint of the 2nd or 3rd leg and into the abdominal cavity. About an 1/8th of a milliliter of hemolymph was collected and deposited into small centrifuge tubes. Initially, we centrifuged the samples for 10 seconds; however, we soon realized that the force from the centrifuge caused the hemocytes to break. Hemolymph samples were then stained with methylene blue and placed in a hemocytometer under a microscope to look for and count hemocytes. Hemolymph samples from

weeks 1 and 2 were combined with 15 µl of the anticoagulant ethylenediaminetetraacetic acid (EDTA) so the samples would remain usable throughout the experiment. Osmolality was determined using an osmometer and the extracted hemolymph.

In an attempt to minimize mortalities, the container with the piece of copper had a water change every other day. As the copper dissipated, the copper ions drastically changed the color of the water, which further stressed the crabs. We expected this not to influence our results, as the copper piece was always present and thus consistently dissolving and releasing ions. Since lead does not dissolve, we did not do water changes for the lead treatment group.

Statistical analyses included grouped one-way ANOVAs comparing treatment groups to total hemocyte count and final osmolality. Average righting time was calculated for each treatment group over the three weeks, and graphed for comparison. Osmolality per treatment group was plotted in a boxplot to display the averages and quartiles.

Results

Data recorded were compiled in Tables 1 and 2, with the first table containing general measurements including sex, weight, carapace length, and width, and the second table including data collected on righting time, osmolality, and hemocyte count. Average righting times per treatment group for each week were calculated and plotted with their standard deviation error bars. The one-way ANOVA comparing the treatment groups to THC resulted in a p-value = 0.3587. The one-way ANOVA comparing the treatment groups to their final osmolalities resulted in a p-value = 0.0633. Since the objective of this experiment was to compare heavy metals to control groups, we also ran one-way ANOVA comparisons of lead and copper to the control separately. This led to a statistically significant p-value of 0.0065 when final osmolalities of the control and lead groups were tested, but not when the control and copper groups were compared.

There were three mortalities during this experiment, with one crab in each treatment group dying halfway to week 3. However, none of these crabs ever had their hemolymph extracted, so an explanation for their deaths is unknown.

Figures and Tables

Table 1: Table containing morphological data of each crab, including what treatment tank they were placed in, their sex, weight, and carapace length and width.

Treatment Group	Crab Number	Sex	Weight (g)	Carapace Length (cm)	Carapace Width (cm)
Control	17	Female	8.62	2.3	2.5
Control	32	Male	7.1	2.3	2.5
Control	100	Male	5.35	2	2.3
Control	1	Male	6.77	2.1	2.3

Lead	19	Male	4.72	1.7	2
Lead	15	Female	5.4	1.8	2
Lead	97	Female	5.58	2.1	2.4
Lead	2	Male	6.94	1.9	2.3
Copper	18	Male	5.55	1.8	2
Copper	17	Male	4.59	1.8	1.9
Copper	3	Male	5.59	1.7	1.8
Copper	99	Female	4.43	1.9	2.3

Table 2: A table containing the recorded data of each crab for righting time, osmolality, and hemocyte count. The crabs that died were numbers 100, 2, and 99.

Treatment Group	Crab Number	Sex	Initial Righting time (s)	Week 2 Righting time (s)	Final Righting time (s)	Week 2 Osmolality (mmol/kg)	Final Osmolality (mmol/kg)	Hemocyte Count (in grid)	Hemocyte Count (out of grid)	Total Hemocyte Count
Control	17	Female	2.31	0.98	2.29	687	940	0	0	0
Control	32	Male	2	0.59	1.98	1035	890	0	0	0
Control	100	Male	1.36	N/A	N/A	N/A	N/A	N/A	N/A	0
Control	1	Male	0.66	N/A	2.79	N/A	942	0	0	0
Lead	19	Male	0.5	3.56	11.35	800	1025	1	0	1
Lead	15	Female	5.08	14.48	16.51	778	1018	2	0	2
Lead	97	Female	4.35	N/A	0.78	N/A	1007	0	0	0
Lead	2	Male	1.8	N/A	N/A	N/A	N/A	N/A	N/A	0
Copper	18	Male	0.81	2.73	2.56	736	1047	0	0	0
Copper	17	Male	0.6	0.75	0.54	877	965	2	0	2
Copper	3	Male	0.91	N/A	0.74	N/A	934	4	12	16
Copper	99	Female	13.94	N/A	N/A	N/A	N/A	N/A	N/A	0

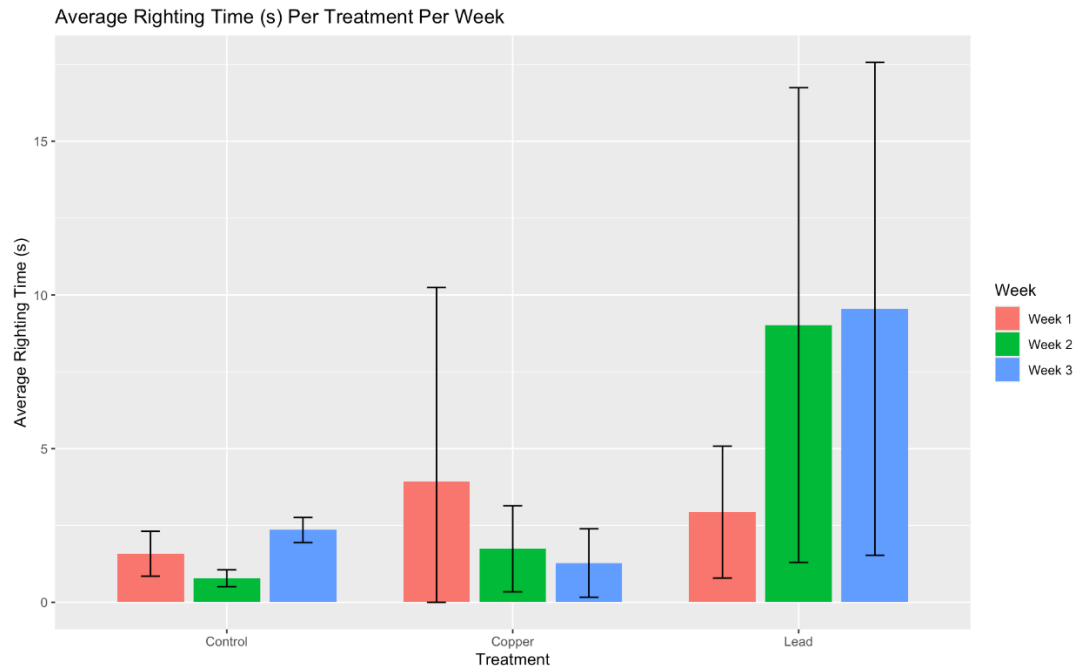


Figure 1: Bar graph depicting average righting time (s) per treatment (Control, Copper, Lead) per week (Weeks 1, 2, and 3). Stable righting time in the control group with little standard deviation. Individuals in the lead treatment saw a drastic increase in righting time from week 1 to week 2, but not as much from week 2 to week 3. Individuals in the copper treatment group saw a decrease in righting time throughout the experiment.

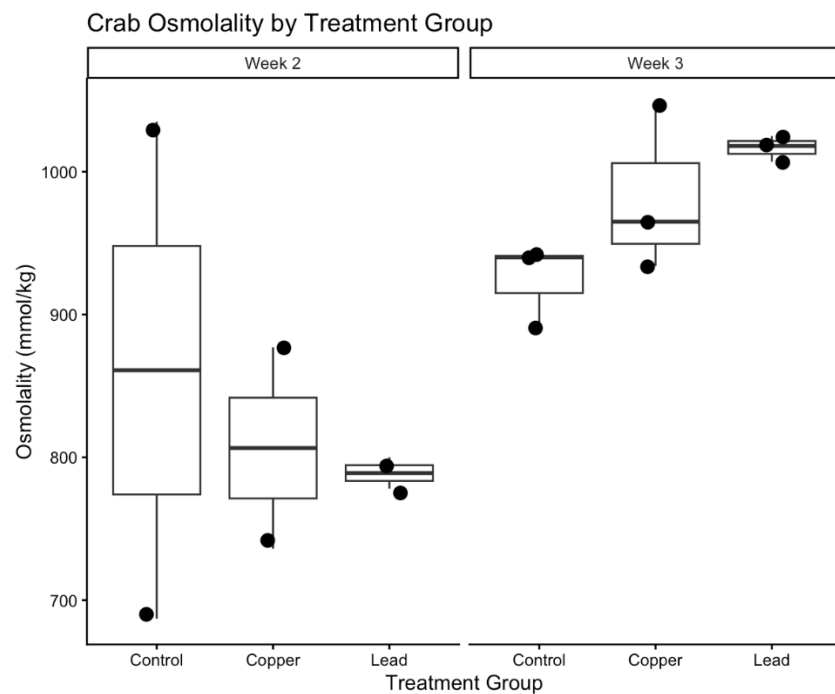


Figure 2: Osmolality per treatment group is compared from week 2 and week 3 and plotted in a boxplot. The lead and copper treatment groups saw a significant increase in osmolality during

this time. When comparing all means in a grouped one-way ANOVA, it resulted in a p-value of 0.0633. However, a grouped one-way ANOVA of the final osmolalities of the control and lead groups produced a p-value of 0.0065, but the comparison of the control and copper groups was not significant.

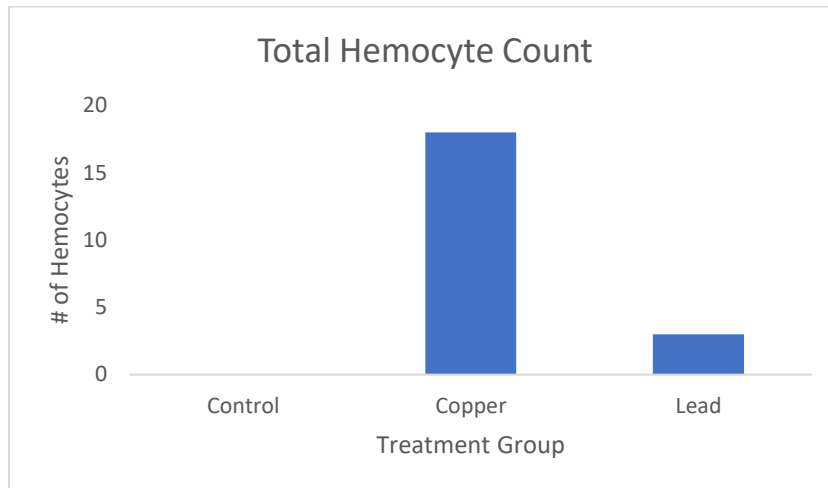


Figure 3: Total hemocyte count (THC) for each treatment group plotted in a bar graph. The control group had no hemocytes present, indicating no immunological stress. While the crabs in the lead group had a few hemocytes detected, the crabs in the copper group had much more. However, through a grouped one-way ANOVA test of all treatment groups, the p-value of 0.3587 is not statistically significant.

Discussion

The goal of this experiment was to determine whether exposure to copper and lead affected the physiology of *Hemigrapsus nudus* by altering righting time, osmolality, and total hemocyte count. Overall, the results suggest that lead exposure had a measurable effect on osmoregulation, while effects on THC were not statistically significant and righting time did not show a clear treatment-related pattern. These findings partially support the hypotheses which stated that heavy metals create sublethal stress in *H. nudus*, but they also indicate that the physiological response may be more complicated than expected.

The most notable result was the significant difference in final osmolality between the control and lead groups. This suggests that lead exposure disrupted the crabs' ability to maintain their internal osmotic balance. High osmolality values indicate a lack of water or high amounts of solutes, and in this case, high amounts of lead. Previous research has shown that heavy metals can interfere with osmoregulatory processes by altering membrane transport and Na⁺/K⁺-ATPase activity, which supports the idea that lead exposure may have impaired ion regulation (Lignot et al. 2000; Rainbow 2002). While the copper group did see an increase in osmolality, the fact that it was not statistically significant may indicate that copper exposure was less disruptive, or that the crabs were able to compensate more effectively for copper than for lead, possibly due to copper being a required ion for oxygen binding in hemolymph. It is also possible

that the water changes for the copper treatment impacted the exposure's intensity, and thus impacted osmolality values to be lower than anticipated.

The THC results suggest that hemocyte abundance did not significantly change across treatments, at least not enough to be detected with the sample size used here. This could mean that the crabs were not exposed to a sufficient metal concentration or duration to produce a measurable immune response. In crustaceans, THC can shift in either direction depending on stress intensity, exposure time, and the balance between immune stimulation and cellular damage (Yildiz and Atar 2002). Righting time also did not statistically distinguish the treatment groups, but as this measure can be easily influenced by body size, sex, handling stress, and individual condition, it may have led to skewed results. It is also important to note that because the crabs were not fed for the duration of the experiment, general energy limitation may have affected activity levels across all groups.

We ran into multiple limitations during this experiment, from our small sample size to damaged hemocytes from putting it through a centrifuge and difficulty identifying hemocytes. Future work should include more replicates to support statistical significance, a larger sample size, and a way to quantify the amount of trace metals in the water since metal toxicity often depends on the dose.

Conclusion

This experiment provided some evidence that sub-lethal lead exposure can disrupt osmoregulation in *Hemigrapsus nudus*, while effects on hemocyte count and righting time were not statistically identified. These results support the idea that heavy metals can produce sublethal physiological stress in intertidal crabs, and that further testing is required. With improved exposure control and a larger sample size, future studies could better determine how copper and lead affect immune and osmotic functions in this species, and eventually relate it to other intertidal crustaceans.

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